Software Requirements Specification for StoryLab Voice: A Voice-Guided Educational Ecosystem with AI Assistant, Multi-Subject Stories, and Virtual Labs

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***Abstract*—This document presents the complete Software Requirements Specification (SRS) for *StoryLab Voice* (Release 1.0), structured in strict alignment with the standard IEEE Std 830-1998 SRS template and ISO/IEC 25010:2011 quality guidelines. StoryLab Voice is an offline-first, voice-guided educational platform developed for resource-constrained rural schools where physical laboratory facilities and reliable broadband connectivity are unavailable. Centered around an empathetic AI Voice Assistant Chatbot, the platform drives two primary pedagogical modules: *StoryLearn* (an interactive, multi-subject storytelling engine across Math, Science, Social Studies, and EVS) and *LabInBox* (a 2D voice-guided virtual science experiment simulator). This specification formally delineates product scope, external interface requirements, detailed system features with priority ratings, and non-functional constraints. Furthermore, it provides a comprehensive process model identification, selecting and analyzing the Spiral Model as the optimal risk-driven development methodology.**

***Index Terms*—Software Requirements Specification (SRS), IEEE Std 830-1998, Conversational AI, Voice-Guided Learning, Virtual Laboratories, Offline-First Architecture, Spiral Process Model, Risk Analysis.**

I. Introduction

A. Purpose

This Software Requirements Specification (SRS) document defines the complete functional, non-functional, and interface requirements for **StoryLab Voice** (Release 1.0). The scope covers an integrated web/mobile application powered by a central AI Voice Assistant Chatbot, driving two primary pedagogical learning modules:

- **LabInBox:** An interactive, browser-based virtual science experiment simulator guided entirely by the conversational voice chatbot.
- **StoryLearn:** A multi-subject story-based learning engine across Mathematics, Science, Social Studies, and Environmental Studies (EVS), narrated and assessed through the conversational chatbot interface.

This document outlines system behavior, interface specifications, performance constraints, and architectural boundaries for full-stack implementation.

Release 1.0 — Software Requirements Specification (SRS) Document and Process Model Analysis adhering to IEEE Std 830-1998 and ISO/IEC 25010:2011 standards.

B. Document Conventions

- **Priority Levels:** Requirements are categorized with **[High]**, **[Medium]**, or **[Low]** priority tags. Child requirements inherit priority from parent modules unless explicitly specified.
- **Typography: Bold text** denotes system roles, core modules, and API calls. `Monospaced text` designates code entities, database schema attributes, and configuration variables.
- **Requirement Identifiers:** Functional requirements are tagged uniquely as `FR-CB-xx` (Chatbot Core), `FR-SL-xx` (StoryLearn), and `FR-LB-xx` (LabInBox).

C. Intended Audience and Reading Suggestions

This document is targeted towards the following project stakeholders:

- **Software Engineers & Developers:** Focus on Section II (Overall Description), Section III (External Interfaces), and Section IV (System Features) for implementation architecture.
- **UI/UX Designers:** Refer to Section II-D, Section III-A, and Section V for conversational UI and accessibility constraints.
- **System Architects:** Focus on chatbot conversation state management, offline voice pipelines, and virtual lab canvas integration.
- **Quality Assurance (QA) Testers:** Review Section IV and Section V to construct test suites and verify acceptance criteria.

Suggested Sequence: Read Section I-D for strategic context, proceed to Section II for product architecture, and examine Section IV for detailed functional requirements.

D. Product Scope

StoryLab Voice is an offline-first educational platform engineered for rural schools. Rather than requiring students to navigate complex graphical menus or parse dense textual material, students interact directly with an empathetic AI Voice Chatbot Companion. The chatbot serves as an always-available personal tutor: narrating localized educational stories, posing guiding questions, answering spontaneous queries in real time, and guiding students step-by-step through virtual science experiments.

Primary Objectives & Goals:

1) Provide a conversational, judgment-free voice interface serving as an always-available personal tutor for rural students.
2) Eliminate physical laboratory equipment dependencies by offering interactive virtual simulations directed entirely via voice dialogue.
3) Contextualize multi-subject syllabus curricula into localized audio stories orchestrated by the chatbot engine.

E. References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Computer Society.
- ISO/IEC 25010:2011, *Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE)*.
- W3C Standard, *WebRTC API & Web Audio API Specifications*.
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II. Overall Description

A. Product Perspective

StoryLab Voice is architected around a central Conversational AI Chatbot layer. The chatbot serves as the primary user interface, coordinating real-time bidirectional data flow between local offline speech models, the LabInBox simulation canvas, and the StoryLearn curriculum database.

B. Product Functions

The principal functional capabilities provided by StoryLab Voice include:

- **Continuous Voice Interaction:** Real-time speech recognition (STT) and local voice synthesis (TTS) for hands-free pedagogical engagement.
- **Contextual Academic Tutoring:** Answering curriculum queries dynamically in simplified vernacular or English.
- **Interactive Story-Based Learning:** Branching educational narrative delivery across four academic subjects (Math, Science, Social Studies, EVS).

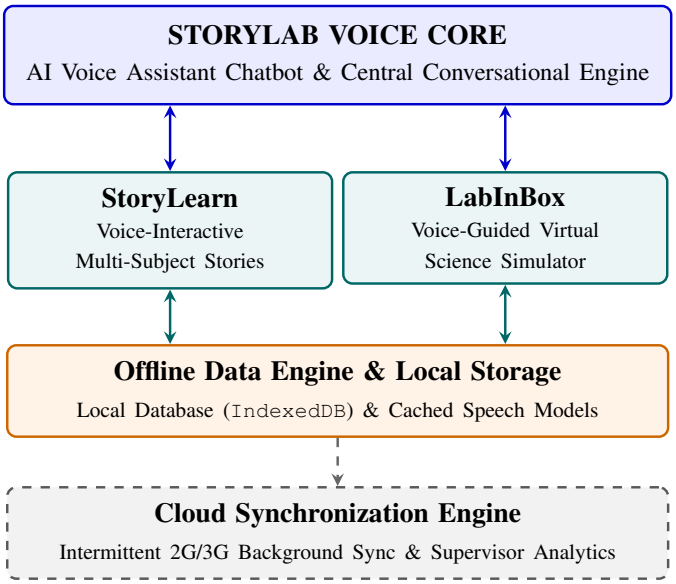


Fig. 1. StoryLab Voice System Architecture and Data Flow.

- **Voice-Controlled Science Experimentation:** Direct speech manipulation of apparatus, chemical mixing, and physics simulations on a 2D HTML5 canvas.
- **Autonomous Offline Operation & Sync:** Zero-connectivity classroom execution with automated background data sync upon reconnecting.

C. User Classes and Characteristics

- **Rural Students (Ages 6–15):** Primary end-users who interact by speaking to and listening to the chatbot companion in their native vernacular language or simplified English.
- **Rural Teachers / NGO Facilitators:** Supervisors who monitor conversation summaries, assign syllabus modules/experiments, and track learning analytics through a dedicated dashboard.
- **System Administrators:** Technical staff who manage regional voice synthesis models, maintain curriculum mappings, and audit AI response accuracy.

D. Operating Environment

- **Client Hardware:** Android 8.0+ smartphones and tablets equipped with a minimum of 2 GB RAM, integrated microphone, and audio speaker hardware.
- **Supported Browsers:** Chromium-based browsers (Google Chrome 88+, Microsoft Edge), and Progressive Web App (PWA) runtime containers.
- **Network Constraints:** Full functionality in disconnected offline settings using lightweight on-device speech synthesis, accompanied by periodic analytics synchronization over intermittent 2G/3G cellular links.

E. Design and Implementation Constraints

- **Storage Limits:** Total local footprint (PWA assets, audio caches, and lightweight acoustic models) must strictly remain ≤ 25 MB.

- **Latency Budget:** Total conversational speech-to-response loop must execute in ≤ 1.8 s on 2 GB RAM hardware.
- **Web Standards:** Application must strictly rely on open web standards (W3C Web Audio, WebRTC, Service Workers, `IndexedDB`).

F. User Documentation

User documentation includes a spoken interactive introductory tutorial delivered by the chatbot on first launch, a simplified visual onboarding card for students, and a web-based supervisor manual for teachers.

G. Assumptions and Dependencies

- It is assumed that client devices possess functional microphone and speaker hardware.
- Speech recognition accuracy assumes reasonable classroom noise levels (≤ 65 dB).
- Dependency on browser support for `IndexedDB` and HTML5 Canvas API.

III. External Interface Requirements

A. User Interfaces

- **Conversational Voice Interface:** Primary bidirectional voice interaction loop enabling 100% hands-free navigation.
- **Visual Avatar / Waveform:** Animated visual feedback indicating real-time chatbot states: *Listening*, *Thinking*, and *Speaking*.
- **Lab Canvas UI:** Clean 2D graphical apparatus display responsive to touch and voice triggers.

B. Hardware Interfaces

- Standard 3.5 mm audio jack / USB-C / built-in microphone for voice capture (16 kHz mono sampling).
- Built-in loudspeaker or external earphones for TTS audio playback.

C. Software Interfaces

- **Local Storage API:** HTML5 `IndexedDB` for storing lesson state, student progress, and local voice response caches.
- **Audio APIs:** W3C Web Audio API and WebRTC for low-latency audio capture and waveform visualization.
- **Service Worker Engine:** CacheStorage API for offline PWA asset caching.

D. Communications Interfaces

- **Sync Protocol:** RESTful HTTPS with compressed JSON payloads.
- **Network Resilience:** Background Sync API handling intermittent 2G/3G connectivity without blocking UI.

IV. System Features & Functional Requirements

A. AI Voice Assistant Chatbot Core

Description & Priority: A bidirectional conversational engine serving as the core pedagogical interface. Priority: **High**.

Stimulus/Response Sequence: Student speaks $\rightarrow$ Speech is converted to text (STT) $\rightarrow$ Contextual dialogue engine parses intent $\rightarrow$ Audio response is synthesized (TTS) and played $\rightarrow$ Avatar waveform displays speaking state.

TABLE I
Functional Requirements: AI Voice Assistant Chatbot

Req ID	Priority	Requirement Statement
FR-CB-01	**High**	The chatbot shall support continuous voice-in/voice-out conversation using Speech-to-Text (STT) and Text-to-Speech (TTS) pipelines.
FR-CB-02	**High**	The chatbot shall answer student-asked educational questions on demand using localized, age-appropriate language.
FR-CB-03	**High**	The chatbot shall detect student confusion or long pauses during activities and proactively offer spoken hints.
FR-CB-04	**Medium**	The chatbot shall provide gentle, supportive feedback on student answers at the end of a topic without interrupting flow.

B. StoryLearn (Voice-Interactive Multi-Subject Stories)

Description & Priority: Delivers syllabus concepts across Mathematics, Science, Social Studies, and EVS through interactive branching stories. Priority: **High**.

TABLE II
Functional Requirements: StoryLearn Module

Req ID	Priority	Requirement Statement
FR-SL-01	**High**	The chatbot shall narrate multi-subject educational stories incorporating local rural variables (crops, markets, festivals).
FR-SL-02	**High**	The chatbot shall pause narrative delivery at key learning moments to ask open-ended or choice-based questions to the student via voice.
FR-SL-03	**Medium**	The system shall dynamically adapt narrative difficulty based on student voice responses and comprehension levels.

C. LabInBox (Voice-Guided Virtual Science Simulator)

Description & Priority: An interactive 2D virtual laboratory environment wherein the AI chatbot acts as a laboratory instructor. Priority: **High**.

V. Non-Functional Requirements

A. Performance Requirements

- **Conversational Latency:** Chatbot speech-to-response generation latency must remain under 1.8 s to maintain natural conversational pacing.

TABLE III
FUNCTIONAL REQUIREMENTS: LABINBOX MODULE

Req ID	Priority	Requirement Statement
FR-LB-01	**High**	The chatbot shall verbally explain lab setup, safety procedures, and step-by-step instructions (e.g., "Now, tell me which solution we should heat first").
FR-LB-02	**High**	Students shall be able to control canvas elements using voice commands spoken to the chatbot (e.g., "Pour the blue beaker into the flask").
FR-LB-03	**Medium**	Upon experiment completion, the chatbot shall discuss findings with the student and generate an automated digital lab report.

- **Offline Capacity:** Core story modules, lab instructions, and cached chatbot responses must function 100% offline without internet.
- **Resource Footprint:** Total application bundle footprint, including lightweight local voice synthesis assets, must not exceed 25 MB.

B. Usability & Accessibility Requirements

- **Conversational UX:** Students must be able to complete 100% of designated learning pathways entirely through voice dialogue.
- **Visual Voice Feedback:** The graphical interface must feature an animated visual avatar/waveform displaying explicit states when the chatbot is thinking, speaking, or listening.

C. Security & Safety Requirements

- **Voice Data Privacy:** Student voice recordings must be processed locally on-device and not permanently stored or transmitted without explicit teacher authorization.
- **Secure Synchronization:** Analytics data synchronized to cloud servers must be encrypted in transit using TLS 1.3.
- **Child Safety:** AI responses must be strictly filtered to guarantee age-appropriate, positive pedagogical interactions.

D. Software Quality Attributes

- **Robustness:** Gracefully handle background classroom noise and diverse regional student dialects.
- **Portability:** Uniform execution across Android smartphones, tablets, and low-cost PWA client browsers.
- **Maintainability:** Modular separation between the voice core, StoryLearn curriculum, and LabInBox simulation canvas.

VI. PROCESS MODEL IDENTIFICATION & ANALYSIS

A. Introduction

StoryLab Voice is an offline-first educational platform designed specifically for rural schools. It combines an AI Voice Assistant Chatbot, StoryLearn for interactive multi-subject stories, and LabInBox for voice-guided virtual science experiments.

The engineering of this platform encompasses several technically challenging and interdependent components:

- On-device Speech-to-Text (STT) and Text-to-Speech (TTS) processing under tight hardware limits.
- Conversational AI dialogue management and contextual state tracking.
- Robust offline functionality with asynchronous, bidirectional cloud data synchronization.
- Real-time voice-controlled 2D virtual physics and chemistry experiment simulations.
- Dynamic and adaptive pedagogical learning algorithms tailored to varied student comprehension levels.

Consequently, the chosen software development life cycle (SDLC) process model must facilitate incremental delivery while systematically identifying, evaluating, and mitigating technical risks.

B. Identified Process Model: The Spiral Model

Identified Process Model: Spiral Model (Boehm, 1988).

The **Spiral Model** is identified as the most suitable software process model for StoryLab Voice. It uniquely merges the iterative nature of rapid prototyping with the controlled, systematic aspects of linear development models, while anchoring every iteration around *explicit risk analysis*. The system evolves through concentric spiral cycles, where each loop progressively elaborates requirements, analyzes risks, builds prototypes, and validates deliverables with end users.

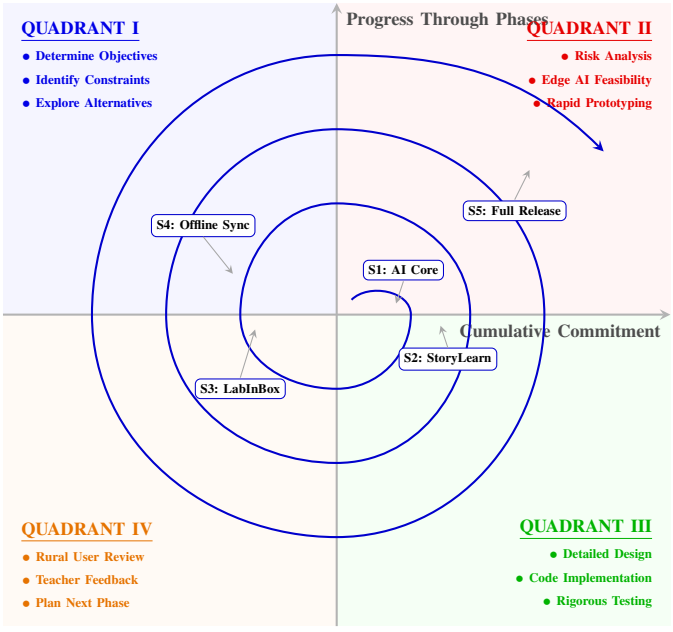


Fig. 2. Four Quadrants of the Spiral Model Tailored for StoryLab Voice.

C. Justification and Risk Analysis

The adoption of the Spiral Model for StoryLab Voice is justified across six foundational engineering dimensions:

1) **High Technical Risk Management:** StoryLab Voice integrates complex edge AI voice interaction, offline speech

inference, dynamic conversational logic, and real-time virtual laboratory simulations on low-cost hardware. The Spiral Model ensures that high-risk technical bottlenecks (such as audio latency and memory overflows) are isolated, analyzed, and mitigated in early spiral loops prior to full-scale engineering.

2) **Prototyping & Incremental Feasibility:** Rapid prototyping is embedded into each cycle. A standalone functional prototype of the AI Voice Assistant Chatbot Core can be built, evaluated on low-end target hardware, and benchmarked before committing development resources to the complex virtual laboratory canvas.
3) **Structured Incremental Development Phases:** The system is engineered logically across five sequential spiral iterations:
   - **Phase 1:** *AI Voice Assistant Chatbot Core:* Build STT/TTS pipelines, voice activity detection, and conversational state tracking.
   - **Phase 2:** *StoryLearn Multi-Subject Story Engine:* Implement branching narratives, dynamic audio streaming, and localized cultural variables.
   - **Phase 3:** *LabInBox Virtual Science Simulator:* Develop HTML5/Canvas 2D simulation physics and map spoken intents to canvas interactions.
   - **Phase 4:** *Offline Storage, Local Caching & Cloud Synchronization:* Implement `IndexedDB` caching, conflict resolution, and background syncing over 2G/3G networks.
   - **Phase 5:** *End-to-End System Integration & Field Testing:* Execute full-system integration, regression validation, performance profiling, and usability trials in rural classrooms.
4) **Continuous User Feedback Loops:** Rural students and educators evaluate each spiral release. This regular feedback loop allows developers to iteratively calibrate speech recognition accuracy for regional accents, refine pedagogical dialogue tones, and enhance accessibility for non-technical users.
5) **Adaptability to Changing Requirements:** Elements such as regional language dialects, state syllabus curriculum revisions, and ambient classroom noise profiles will inevitably evolve during field trials. The Spiral Model effortlessly accommodates modifications at each planning boundary.
6) **Early Testing & Multi-Stage Validation:** Every spiral iteration culminates in rigorous formal verification. Performance defects—such as latency spikes (> 1.8 s), speech recognition mismatches, and animation rendering lags—are caught and resolved early in the lifecycle rather than during final deployment.

D. Conclusion on Process Model Selection

The Spiral Model is the optimal software development process model for StoryLab Voice because it provides a risk-driven, iterative, and flexible framework. It establishes the exact structural rigor necessary for a multi-faceted educational platform integrating conversational AI, voice processing, offline execution, and interactive scientific simulations. By coupling prototype-driven risk analysis with continuous field evaluation, the engineering team guarantees high recognition accuracy, minimal response latency, and maximum pedagogical efficacy prior to final release.

TABLE IV
COMPARISON OF CANDIDATE SDLC PROCESS MODELS

Criterion	Waterfall	Standard Agile	Spiral Model
Risk Handling	Poor	Moderate	**High / Explicit**
Edge AI Prototyping	No	Moderate	**Native**
Requirement Flexibility	Very Low	High	**High**
Hardware Constraints	Risky	Reactive	**Proactive**
Suitability	Not Suitable	Partial	**Optimal**

VII. CONCLUSION

This Software Requirements Specification has articulated the complete functional, non-functional, and structural requirements for Release 1.0 of *StoryLab Voice*, following IEEE Std 830-1998 guidelines. By situating an AI Voice Assistant Chatbot at the center of both multi-subject narrative learning (*StoryLearn*) and hands-on virtual laboratory experimentation (*LabInBox*), the platform effectively democratizes quality STEM and humanities education for resource-constrained rural schools. The formal selection of the **Spiral Model** provides a disciplined, risk-mitigated pathway toward successful development, rigorous testing, and field deployment.

APPENDIX

- **AI:** Artificial Intelligence.
- **COPPA:** Children's Online Privacy Protection Act.
- **EVS:** Environmental Studies (Elementary School Curriculum).
- **IndexedDB:** Client-side transactional NoSQL database in modern web browsers.
- **PWA:** Progressive Web Application.
- **SDLC:** Software Development Life Cycle.
- **SRS:** Software Requirements Specification.
- **STT:** Speech-to-Text (Voice Recognition).
- **TTS:** Text-to-Speech (Speech Synthesis).
- **WebRTC:** Web Real-Time Communication API.

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